

CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN – International Conference on Project MANagement / HCist – International Conference on Health and Social Care Information Systems and Technologies 2024

Evaluating situation-aware user interface adaptations for e-commerce applications

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Abstract

Adaptive user interfaces can lead to better user experience and task accomplishment, when tailored to the users' needs and helping to understand the complex task structure and functionality of e-commerce applications. SitAdapt is an intelligent development and control system for building and running situation-aware adaptive user interfaces for model-based e-commerce web-applications. This paper first briefly introduces the concept of adaptive user interfaces and the underlying SitAdapt system for developing and running such interactive applications. It then focuses on the results of a study-based evaluation of users' familiarity with the term adaptivity in general and the different adaptation-types implemented for an interactive travel-booking application that serves as a testbed for assessing the performance and user experience of situation-aware web-applications. The main research contribution of this paper is to examine how users in different age groups and with different experience in digital technologies understand, perceive and accept the concepts of user interface adaptivity.

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Peer-review under responsibility of the scientific committee of the CENTERIS - International Conference on ENTERprise Information Systems / ProjMAN - International Conference on Project MANagement / HCist - International Conference on Health and Social Care Information Systems and Technologies

Keywords: Adaptive user interface; situation-awareness; e-commerce web-applications; user experience evaluation

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1. Introduction and related work

SitAdapt [1] is a development framework and control system for building and running situation-aware interactive applications with runtime-adaptive user interfaces. We have focused our research on adaptive e-commerce web applications. A travel-booking portal serves as a testbench for prototyping and evaluating user interface adaptivity. In the first section of this paper, we give an overview of the various types of adaptive user interfaces and a brief introduction to the features of the SitAdapt system.

The effects of user interface adaptations and the concept of adaptivity in general are not easy to grasp for typical users of e-commerce applications. Therefore, we carried out a questionnaire-based user study to assess users' familiarity with the term adaptivity and adaptive user interfaces. This study is discussed in the second section. As an example of an adaptive interactive system, we chose the travel-booking portal as a typical web-based e-commerce application. The last section concludes the paper, points out limitations and identifies future research opportunities.

1.1. Types of adaptive user interfaces

Adding adaptivity to user interfaces of e-commerce web applications can have positive effects on the usability and the understandability of such applications. Adaptivity can make the functionality and task structure of the systems more transparent and raise user experience all through the customer journey. Adaptivity can be implemented in many ways [2]. When focusing on user interface adaptivity, typically the following four types of adaptation can be distinguished [3], [4]:

- **Adaptable user interfaces.** The user interface is a-priori customized to the personal preferences of a user. Customization can be directly applied by the user at the beginning of a session. The user then must be aware of the possible adaptations. This can, e.g., be done by offering a menu-based dialogue for activating adaptation options, or by introducing a chatbot that leads the user through the set of possible user interface adaptations. The chosen adaptations will remain active for a registered user throughout the session and in future sessions as long as not modified by the user.
- **Persona-related adaptive user interfaces.** In this case a user must be mapped to a specific persona before persona-related adaptations are activated. For e-commerce environments user personas are often modeled based on marketing knowledge and customer studies for allowing business applications to react to the preferences of different customer groups. A registered and recurring customer can then be mapped to one of the available personas. However, it is also possible to generate new personas by analyzing the user behavior in the different phases of the digital customer journey and by mining typical behavioral patterns [5]. From a set of such patterns, a new persona can then be derived. Before persona-specific adaptations are activated users should be informed about the reasons for and the effects of these adaptations with the option of disabling some or all of them.
- **Semi-automated adaptive user interfaces.** The user interface provides recommendations for adaptations, which can be accepted by the user or not. This can be accomplished by intelligent systems that observe the user's interactions and the user's cognitive and emotional state during a session [6], [7]. User behavior can then be mapped to the task and business models of the application and, e.g., in case of significant deviations from the workflow suggested by the models, or, if a critical threshold of cognitive load or emotional arousal has been exceeded, a recommendation can be given that lets the user return to a more promising or comfortable work context.
- **Automated adaptive user interfaces.** Applications that implement this type of adaptive user interfaces react in real-time and automatically to the current situation of a user during a session [8]. As with semi-automated adaptivity, the user is observed by the system, the user behavior is evaluated and the system decides, if a situation has occurred that requires an adaptation to optimize the work context and/or improve the user experience. The idea behind this approach is to provide a more intelligent way of user-system-dialogue. This approach can be implemented, e.g., by using situation rules that trigger real-time adaptations. The quality of the rules is critical to the usefulness of such systems. To raise the quality of the adaptive behavior and the generated adaptations over time, a machine learning component could be added to such systems. A promising direction could be the usage of reinforcement learning [9]. When using such intelligent adaptive systems, the user should always be able to learn about their adaptation capabilities, before and during sessions with the application.

With respect to temporal aspects of adaptations, one must distinguish between:

- A-priori adaptations that are prepared before the actual session begins, but when the user can already be observed and interact with the chosen platform. Such adaptations could, e.g., concern user-specific usability topics and are activated when the application starts.
- Adaptations that are generated and activated at runtime and might cover both, domain-dependent and user-specific topics.
- A-posteriori adaptations based on conclusions drawn from observation data from one or more user-sessions that already have terminated. Such adaptations might, e.g., be able to provide better UX and user support in situations that have caused problems for some users and could not be resolved by the already implemented adaptations. Such newly created adaptations can either be implemented by human developers and be integrated into new versions of the interactive system, or be generated automatically by the system, using the knowledge inferred from the observation data and available resources of a model-based user interface development system (MBUID).

1.2. SitAdapt overview

The SitAdapt system uses the model based PaMGIS framework [10] as a resource base and offers a situation analysis and rule model for development and runtime support for adaptive user interfaces. It can react to context changes in the user's environment, the user's behavior, and emotional state in real-time. To support the different types of adaptations, the observer part of the system collects data from different kinds of inputs (eye-tracker, wristband, facial expression recognition software interface, and metadata from the application regarding the user and the environment).

The construction of adaptive interactive systems can be accomplished by carefully modeling all the required business processes and related interactive tasks and by user involvement during software development [11].

When using a model-based user interface development approach in the web application development process as we did with SitAdapt, the models, relationships, and mappings between the various models and between the models and the final user interface are still available and accessible by the target software, when the first application prototypes have been implemented. At runtime the implemented applications can access the models and other resources for creating real-time adaptations.

Model-based approaches for building adaptive interactive systems have a long history and appear in many different varieties. A comprehensive overview on model-based approaches for intelligent user interface adaptation is given in [12]. The paper also presents a conceptual reference framework for defining the structure and architecture of software systems that support adaptive UIs. The new framework extends the CAMELEON reference framework [13] for model-based user interface development environments with components that are necessary for building and running systems with adaptive user interfaces. Our own SitAdapt system interacts with the PaMGIS framework that follows the earlier CAMELEON reference framework for identifying the relevant models and their inter-relations. To allow for modeling adaptivity some new modeling resources had to be added to the PaMGIS framework. For designing real-time adaptations, SitAdapt focuses on the exploitation of rules that can access the dynamically changing observation data that are stored as situation profiles in a database.

2. Studying the Acceptance of User Interface Adaptations

To evaluate the overall SitAdapt approach and at the same time get constructive and well justified user feedback we have carried out a user study as a questionnaire-based online survey and a follow-up experimental scenario-based user study in our lab. This paper only discusses the results of the online survey.

To evaluate the current level of knowledge of the different age groups about adaptive user interfaces for interactive systems, an anonymous online survey was carried out as part of a pretest. A total of 110 test subjects took part in the period from December 7th, 2023, to January 18th, 2024. Most of the subjects were recruited using the mailing lists for students, non-technical staff, technical staff, and scientific staff of the different faculties of TH Augsburg. The

results of the pretest were incorporated into the usability lab test in the next step. The survey design used primary research as an empirical method to collect new, previously unavailable data. These include, among other things, the connection between technology affinity and the use of adaptable and adaptive systems¹.

In the first step, to determine whether there is a significant relationship between the affinity for technology and the age of the test subjects, the participants were divided into different age groups based on the different currently perceived generations (see Table 1).

Table 1. Participants spread across different generations.

Age Range	Number of participants	Percentage
14 – 28 years old	68	61.82%
29 – 43 years old	20	18.18%
44 – 58 years old	10	9.09%
59+ years old	12	10.91%

In our test, a majority of 68 participants were in the 14-28-years-old age range. The group of the 29–43-years-old is represented by 20, the group of 44-58-years-old by 10, and the group of 59+ years-old is represented by 12 subjects.

In the second step, the different age groups were asked to provide a self-assessment of their experience with using digital technologies. The answer options were very experienced, experienced, little experienced and not experienced. The breakdown of the respective answers can be found in Table 2.

Table 2. The evaluation of technology affinity according to age groups.

Age Range	Very experienced	Experienced	Little experienced	Not experienced
14 – 28 years old	35 (51%)	32 (47%)	1 (1%)	0
29 – 43 years old	6 (30%)	11 (55%)	3 (15%)	0
44 – 58 years old	4 (40%)	3 (30%)	2 (20%)	1 (10%)
59+ years old	4 (33.33%)	2 (16.67%)	3 (25%)	3 (25%)

Based on the data from Tables 1 and 2, the Pearson chi-square test [14] was performed as a method of multivariate data analysis. This test examines whether there is a connection between the two categorical variables “age group” and “technology affinity” and how strong this is. An $\alpha = 0.05$ was used as the significance level. The calculations were carried out using the JMP statistical software.

This results in a Pearson chi-square coefficient χ^2 of 35.378. Since this value is a non-standardized measure, the contingency coefficient C according to Pearson must be determined accordingly with $C = 0.493$. To exclude the influence of the number of characteristic expressions, the corrected contingency coefficient C_{corr} is determined. The corrected contingency coefficient between the age group and the affinity for technology is 0.569 (56.9%), i.e., there is a moderate connection [15] between the two characteristics which is statistically significant (p -value < 0.0001). The next question in the survey deals with the familiarity of the term “adaptable/adaptive in connection with digital technologies” in the age groups and whether a meaning can be assigned to it. The answer option was yes, or no. Table 3 shows the answers broken down by age group.

¹The questionnaire (including an English translations of the questions) is made available at the end of the website of our research group: <https://www.tha.de/Informatik/HSA-aue.html>

Table 3. Familiarity of the terms adaptable/adaptive by age group.

Age Range	Yes	No
14 – 28 years old	35	33
29 – 43 years old	9	11
44 – 58 years old	3	7
59+ years old	3	19

Based on the answers, the Pearson chi-square test [14] was also carried out here to examine whether there is a relationship between the two categorical variables age group and knowledge of the term adaptable/adaptive and how strong this is. No statistically significant relationship was found in this sample as a p-value = 0.4860 was determined.

Analogous to the different age groups, it was examined whether there is a connection between affinity for technology and awareness of the term “adaptable/adaptive”. For this purpose, the groups “very experienced” were combined with “experienced” as well as “little experienced” and “not experienced”. A merger was necessary because the number of participants per individual group was too small (see Table 4).

Table 4. Familiarity of the terms adaptable/adaptive by technology affinity.

Technology affinity	Experienced	Not Experienced
Yes	50	0
No	47	13

The result is a Pearson chi-square coefficient χ^2 of 12.285. The contingency coefficient according to Pearson C is 0.317, the corrected contingency coefficient between the age group and the affinity for technology is 0.448 (44.8%), there is therefore a moderate connection [15] between the two characteristics which is statistically significant (p-value = 0.0005).

Our pretest showed that there is a connection between the two variables, for technology affinity and age groups. The affinity for technology is higher among the younger age groups than among the older ones. There is also a connection between affinity for technology and whether the concept of adaptable/adaptive is known in connection with digital technologies. Among the technology-savvy participants, 50 out of 97 technology-savvy users knew this term. The term was completely unknown to non-technical users.

The next questions dealt with the different adaptable and adaptive functions and how these are known to the target groups of technology-savvy and non-savvy users. The first of these questions was about awareness of the possibility of using settings menus to adapt the user interface to your needs (browser, office programs, etc.). This is an adaptable function. The answers from the users are shown in table 5.

Table 5. Familiarity of settings menus by technology affinity.

Technology affinity	Experienced	Not Experienced
Yes	82	10
No	15	3

A Pearson chi-square test [14] was also carried out to examine the alternative hypothesis in which there is a connection between the two categorical variables of technology affinity and the familiarity of settings menus. However, no statistically significant relationship was found in the sample as p-value was 0.4860.

The next question was intended to answer whether participants know that advertising on the Internet can be tailored to you, depending on your surfing behavior. See table 6 for the answers. This is an example of a fully automated adaptive function.

Table 6. Familiarity of personalized advertising by technology affinity.

Technology affinity	Experienced	Not Experienced
Yes	86	13
No	11	0

Here too, a Pearson chi-square test [14] was carried out to examine the alternative hypothesis in which there is a relationship between the two categorical variables of technology affinity and the awareness of personalized advertising. However, no statistically significant relationship was found in the sample as p-value was 0.2006.

In the next question, it should be clarified whether the participants are aware that, for example, navigation software automatically switches to dark mode in the dark or at a certain time of day, for creating a display that is easier on the eyes and produces less glare, for example in the car. The answers to this question are shown in table 7.

Table 7. Familiarity with the 'dark mode' by technology affinity.

Technology affinity	Experienced	Not Experienced
Yes	84	6
No	13	7

The alternative theory was put forward that there is a connection between affinity for technology and awareness of dark mode. This results in a Pearson chi-square coefficient χ^2 of 12.605. The contingency coefficient according to Pearson C is 0.321, the corrected contingency coefficient between awareness of dark mode and affinity for technology is 0.454 (45.41%), so there is a moderate connection [15] between the two characteristics that is statistically significant (p value is 0.0004).

The next question was intended to determine whether there is a connection between the user's affinity for technology and the awareness of responsive web design, in which the layout and, if necessary, content adapt to the respective device. Answers of the users can be found in table 8.

Table 8. Familiarity with responsive web design by technology affinity.

Technology affinity	Experienced	Not Experienced
Yes	86	3
No	11	10

This results in a Pearson chi-square coefficient χ^2 of 31.921. The contingency coefficient according to Pearson C is 0.460, the corrected contingency coefficient between the awareness of responsive web design and the affinity for technology is 0.651 (65.51%), so there is a moderate connection [15] between the two characteristics which is statistically significant (p value < 0.0001).

3. Conclusions and future work

Adaptive interactive systems can serve multiple purposes. For instance they allow experienced users to enjoy the full potential of an application by manually choosing their individual settings, they help all user groups to find a balanced user experience by automatically mapping the settings and functionality of the application to their matching user persona, and, they are able to support unexperienced or new users to accomplish their tasks by leading them through the application, giving them hints for navigation and functionality and automatically adapting the look and feel of the user interface to the individual needs and the current mood of their users.

The SitAdapt system allows to build all types of adaptive interactive systems. It is currently used in a lab environment to study situation-aware interactive systems, user experience with such systems, and gather experience of all kinds of adaptivity and with different user groups. We understand that observing users during their sessions

poses some ethical problems, e.g., when recognizing their emotions, following their gazes, or measuring their cognitive load. Especially, when such technologies leave the lab environment and are integrated into consumer products and services, ethical standards and value-based design guidelines must be respected [16], and users must a priori be informed about the applied monitoring and measuring technologies. Users must also be able to control their privacy level. However, the huge potential of situation-aware and fully adaptive systems can heavily contribute to arrive at better usability, task-management, more safety in technical applications, and better user experience for each individual user, and therefore should be thoroughly and carefully researched for all application areas.

With the advent of more sophisticated AI technologies, e.g., intelligent chatbots or avatars based on large language models (LLMs), such technologies will also play a major role in automated adaptive user interfaces. Currently the hardware industry is presenting their first processors with integrated AI accelerators that can support small or medium sized LLMs directly on a laptop or smartphone without the need of access to AI server clouds². For many application areas, including adaptive user interfaces for the e-commerce sector, this new direction could offer an elegant way to respect users' privacy, because the systems will become powerful enough to execute all necessary AI and monitoring software locally at the workplace.

Our study on the awareness and the concepts of adaptivity shows that the potential of adaptable and adaptive interactive systems is not as well-known as other achievements of information technology. Unexperienced users, who would benefit the most from adaptivity, have little knowledge of adaptive user interfaces, adaptivity in general, and of the underlying concepts. The significance of our survey is limited, because it mostly reached students and staff of our university. We fear that knowledge about adaptivity is even less developed among IT users from the general population. Therefore, we are planning an extended study that is targeted to a wide spectrum of typical IT users. To promote adaptive interactive systems and adaptive technology, the technology itself should and could be used to produce more trustworthy, understandable, and self-explaining user interfaces for individual users with different skill-levels. We will focus our future research on these directions, because we are convinced that better acceptance of adaptive technologies will at the same time raise individual user experience, customer satisfaction, and the business success of the developers and providers of adaptive applications and services.

Acknowledgements. The authors want to thank Torsten Straßer for his statistical advice and Felix Schmidberger for implementing the online questionnaire as part of his bachelor thesis.

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